Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Engineering · Class 11

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Matrices: Types and Addition

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Learning Objectives

• Students will define a matrix and state its order.

• Students will identify the types of a matrix: square, row, column, diagonal and zero.

• Students will add two matrices of the same order correctly.

Key Concepts

• A matrix is a rectangular arrangement of numbers in rows and columns, written inside square brackets.

• The order of a matrix is the number of its rows by the number of its columns, written as m x n.

• A square matrix has the same number of rows and columns, a row matrix has only one row, and a column matrix has only one column.

• A zero matrix has all its entries equal to zero, and a diagonal matrix has all its off-diagonal entries equal to zero.

• Two matrices can be added only when they have the same order, and addition is done entry by entry.

Teaching Methodology (Step by Step)

1. Show a table of marks on the board and ask how the numbers could be written more neatly as one block.

2. Define a matrix, explain rows and columns, and write the order of the marks table.

3. Introduce the main types one by one — square, row, column, diagonal and zero — with a small example of each.

4. Explain the rule for addition: the same order is needed, and entries are added in matching positions.

5. Work out two additions on the board, including one with negative entries.

6. Give students three pairs of matrices to add on their own, then summarise the types and the addition rule.

Classroom Activities

• Type hunt: students sort a set of matrices into square, row, column, diagonal and zero matrices.

• Addition race: in pairs, students add two matrices of the same order and check each other's answers.

• Real-life link: students write one example where a table of numbers, such as marks or prices, can be written as a matrix.

Assessment Questions

1. Define a matrix and state its order for a 3 x 2 matrix.

2. Can a 2 x 3 matrix be added to a 3 x 2 matrix? Give a reason.

3. Add the matrices A = [[2, 5], [1, 4]] and B = [[3, 1], [6, 2]] and state the order of the sum.

Homework

• Write the order of each of the matrices given in the textbook exercise and name their types.

• Add the three pairs of matrices given in the exercise and verify that A + B = B + A in each case.

Time Allocation

40 minutes

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